

MEC - 110 MONEY FINANCIAL INSTITUTIONS & MARKETS



Money, Financial Institutions and Markets

(Volume-1)

School of Social Sciences
Indira Gandhi National Open University
New Delhi

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COURSE INTRODUCTION

MEC 110: Money, Financial Institutions and Markets

The structure of MA Economics programme has been spread over four semesters. In each of the first three semesters, three courses (in the multiple of six or eight credits) have been included in a well thought manner. The two courses i.e., Micro Economic Analysis (MEC-101) and Macro Economic Analysis (MEC-102) provide the theoretical foundation and the knowledge of mathematical and statistical tools in MEC 203, enabling to understand micro and macro theories in analytical manner. Another set of three courses included in semester two lead to further insight into the area of growth and development. The knowledge of economic theory both in micro and macro domain and insight into growth and development prepare the foundation for entering the empirical part and take up the various components of Indian economics policy (MEC-205). Knowledge of theoretical and applied part of economics make a solid ground for making in-depth study of various economic events, developments and situations. However, before taking up in-depth study, the knowledge of conceptual foundation, tools and techniques of research methodology is required. These have been provided in the course Research Methods in Economics (MEC 109).

Having the knowledge of core courses in the first and the second semesters, with the intention to have knowledge of area specific core courses, another set of three courses viz., Public Economics (MEC-106) International Trade and Development (MEC-107) and Money, Financial Institutions and Markets (MEC 110) have been introduced in the third semester.

We have covered issues related to monetary policy in macro-economic analysis, yet the various topics related to monetary economics such as functions of money and its relation and interaction with financial institutions and markets could not be covered, which are otherwise essential to have insight into the functioning of various kinds of markets in the economy. Hence, MEC 110 course as an extension of macro dimension of economics has been included in this semester.

In order to comprehend this course in a meaningful manner, students should have a clear idea of the conceptual foundations and functioning of the financial system. In this process, it is necessary to understand the role and functions of money, and the transmission mechanism of monetary policy.

This course is extremely relevant because money, financial institutions, and markets form the cornerstone of modern economies. They serve as the channels through which resources are allocated, transactions are facilitated, and economic prosperity is achieved. Their roles are multifaceted and intertwined, shaping the economic landscape and impacting the lives of individuals, businesses, and societies around the globe. Thus, understanding functions of money, financial institutions, and markets, interactions, and implications is essential for policymakers, economists, investors, and citizens alike, helping them to navigate the complexities of the financial system.

The course discusses how capital flows through the economy and how financial intermediaries play a crucial role in channelling funds between surplus and deficit units in the economy. It deliberates upon the roles, functions, and regulatory frameworks of money markets, capital markets, bond markets, and derivatives. It also provides insights into market dynamics, participant behaviour, and factors driving asset valuations. The

course also delves into functioning of Commercial banks, non-bank financial intermediaries, and regulatory bodies which form the backbone of the financial system and play an important role in maintaining economic stability. Further, it equips with the understanding of 'Efficient Portfolio Frontier' and asset pricing theories which are essential for investors seeking to optimize returns and manage risk. To understand the challenges and complexities of maintaining a stable financial system, this course also presents the comprehensive understanding of non-performing assets, foreign exchange stability, and behavioural finance.

This course is divided into eight blocks. Each block is interconnected. The structure of the course is as follows:

Block 1: The Financial System is a foundational block. This block has 3 units. It provides an overview of the financial system, elucidating the interlinkages between economic agents, markets, and institutions. It also gives an overview of the process of flow of funds, the role of financial intermediaries, and the fundamentals of basic business accounting, setting the stage for a deeper understanding of Money, financial institutions and markets.

Block 2 entitled **Money** contains three units. This block unravels significance of money in the modern economy. From understanding its functions and liquidity to exploring theories of demand for money and the process of money supply, this block gives insights into Money as a fundamental building block of monetary systems.

In **Block 3: Monetary Policy**, we have discussed the pivotal role that the Central Bank plays in shaping monetary policy and presented their functions, objectives, and instruments. Three units have been covered in this block which explores the transmission mechanisms of monetary policy and analyze its implications on financial institutions and markets.

Block 4: Financial Markets deals with the functioning of the financial market. Its content is spread over four units. The intricacies of money markets, capital markets, bond markets, and derivatives, throwing light on their functions, participants, and dynamics have been discussed in this block.

Block 1

The Financial System: Overview

Unit 1	Economic Agents
Unit 2	Financial Intermediation
Unit 3	Basic Business Accounting

UNIT 1 ECONOMIC AGENTS

Structure

- 1.0 Objectives
- 1.1 Introduction
- 1.2 The Nature of Financial System
 - 1.2.1 Financial Institution
 - 1.2.2 Financial Markets
 - 1.2.3 Financial Instruments
 - 1.2.4 Financial Services
- 1.3 Participants in Financial Markets
- 1.4 Financial Markets
 - 1.4.1 Types of Financial Markets
 - 1.4.2 Importance and Functions of Financial Markets
- 1.5 Financial Instruments
 - 1.5.1 Types of Financial Instruments
- 1.6 Let Us Sum Up
- 1.7 Key Words
- 1.8 Some Useful References
- 1.9 Answers/Hints to Check Your Progress Exercises

1.0 OBJECTIVES

After going through this unit, you will be able to:

- Explain the concept and features of financial system;
- Describe types of financial institutions, financial markets, financial instruments;
- Discuss the functions of financial institutions and markets;
- Highlight the importance of financial institutions; and
- Recognise different financial instruments and analyse the need for regulating financial institutions.

1.1 INTRODUCTION

You must have read in Microeconomics that economic agents are individuals or entities that participate in economic activities by making choices, engaging in transactions, and influencing the allocation of resources in an economy. The main economic agents are households,

firms, government, financial institutions etc. These economic agents interact with each other in markets. They influence demand and supply, prices and overall functioning of the economy. In this unit, we will discuss the main economic agents in the context of financial markets and institutions. For example, households play an important role in financial markets principally as savers and also as borrowers. They save money by depositing it in banks or investing in various financial instruments like savings accounts, bonds, stocks etc. At the same time, they also borrow money for various purposes like buying a house, car etc. Similarly, firms act as borrowers and investors, financial institutions act as intermediaries, and the government as regulator and participant in financial activity in the economy.

The economic agents collectively influence the flow of money, credit, and investments in financial markets. This in turn impacts interest rates, asset prices, and overall economic stability.

Consider if you were to start a business that manufactures special equipment for persons with disabilities which ease their day-to-day functioning, but you have no funds to invest in this special equipment. At the same time, there is Raj who has inherited a huge amount of money and wants to invest it in some business. If you and Raj could get together, both of you could be in a win-win situation where you get to have funds for your business and Raj is able to invest his idle money. In this example, both you and Raj are economic agents. Financial markets (bond and stock markets) and financial intermediaries (banks, insurance companies, and pension funds) have the basic function of getting people such as you and Raj together by moving funds from those who have a surplus of funds (Raj) to those who have a shortage of funds (you).

Financial institutions, financial markets, financial services, financial instruments, financial practices and financial transactions together form **the Financial System**. The term financial institutions refer to all kinds of entities which intermediate and facilitate financial transactions of both individual and corporate clients. Essentially, financial institutions are the intermediaries between the savers and borrowers of the money. They transfer financial resources from the savers to the investors or borrowers. Financial institutions have several functions that assist the economic agent in fulfilling their various needs. They deal in all types of finance-related businesses thus playing a crucial role in the economy.

There are different types of financial institutions: banking and non-banking. The function of these institutions is different and they provide different services and have different levels of risk associated with it. In this unit, we will explore mainly the functions, roles, and types of financial institutions, financial markets and financial instruments.

1.2 THE NATURE OF FINANCIAL SYSTEM

A financial system is a complex, well-integrated set of sub-systems of financial institutions, markets, instruments, and services which facilitate the transfer and allocation of funds, efficiently and effectively between different economic agents.

There are four constituents of a financial system, viz. (i) financial institutions, (ii) financial markets, (iii) financial instruments, and (iv) financial services.

1.2.1 Financial Institutions

Financial institutions are business organisations that act as mobilisers and depositors of savings and as purveyors of credit or finance. They also provide various financial services to the community.

Financial institutions have been variously classified. Two important classifications are as follows:

- A) Banking institutions and non-banking institutions
- B) Intermediaries and non-intermediaries.

A) Banking Institutions and Non-banking institutions

The banking institutions have quite a few things in common with the non-banking ones, but their distinguishing character lies in the fact that, unlike other institutions, (i) they participate in the economy's payments mechanism, i.e., they provide transactions services, (ii) their deposit liabilities constitute a major part of the national money supply, and (iii) they can, as a whole, create deposits or credit which is money. The distinction between the two has been highlighted by Sayers by characterizing the former as “**creators**” of credit, and the latter as “**purveyors**” of credit.

B) Intermediaries and non-intermediaries

Intermediaries intermediate between savers and investors; they lend money as well as mobilise savings; their liabilities are towards the ultimate savers, while their assets are from the investors or borrowers. Non-intermediary institutions are in the lending business but their resources are not directly obtained from the savers. All banking institutions are intermediaries, many non-banking institutions also act as intermediaries and when they do so they are known as **non-banking financial intermediaries**. Some examples of non-banking financial intermediaries include mutual funds, investment funds, insurance companies, and pension funds. These institutions provide a valuable alternative to traditional bank financing.

1.2.2 Financial Markets

Financial markets are a mechanism enabling participants to deal with financial claims. The markets also provide a facility in which their demands and requirements interact to set a price for such claims. The participants on the demand and supply sides of these markets are financial institutions, agents, brokers, dealers, borrowers, lenders, savers, and others who are interlinked by the laws, contracts, covenants and communications networks.

Financial markets have been variously classified. Two important classifications are **A. Primary market and secondary market and B. Money market and Capital market**. We will discuss about financial markets in details in section 1.4.

1.2.3 Financial Instruments

A financial instrument is a claim against a person or an institution for the payment at a future date of a sum of money and/or a periodic payment in the form of interest or dividend. The term 'and/or' implies that either of the payments will be sufficient but both of them may be promised. We will discuss more about financial instruments in section 1.5.

1.2.4 Financial Services

Financial intermediaries provide key financial services such as merchant banking, leasing, hire purchase, credit rating, etc. Financial services rendered by financial intermediaries bridge the gap between lack of knowledge and information on the part of investors and their participation in financial markets and working with increasingly sophisticated financial instruments. Some of the important functions of Financial Services are:

- 1) **Facilitating Transactions:** Enabling the exchange of goods and services within the economy.
- 2) **Mobilizing Savings:** Gathering savings that would otherwise have limited investment opportunities.
- 3) **Allocating Capital:** Directing capital funds to finance productive investments.
- 4) **Monitoring Management:** Ensuring that allocated funds are used as intended.

Financial service institutions provide a diverse range of services to cater to the needs of individual users. Some of them are:

- 1) **Provision of funds:**
 - a) Venture capital
 - b) Banking services
 - c) Asset financing

d) Trade financing

e) Credit cards

2) Managing investible funds:

a) Portfolio management

b) Merchant banking

c) Mutual and pension funds

3) Risk financing:

a) Project preparatory services

b) Insurance

c) Export credit guarantee

4) Consultancy services:

a) Project preparatory services

b) Project appraisal

c) Business advisory services

d) Valuation of investments

e) Credit rating

f) Merger and acquisition.

As discussed above, the financial services provided by these intermediaries close the gap between investors' lack of knowledge and their involvement in financial markets. This helps to facilitate informed decision-making and promotes effective engagement with increasingly sophisticated financial instruments.

1.3 PARTICIPANTS IN FINANCIAL MARKETS

The important participants in financial markets are business firms, investors, brokers, speculators, hedgers, and arbitrageurs. Firms use stock and bond markets to raise capital from investors; speculators look up to various asset classes to make directional bets on future prices; hedgers use derivatives markets to mitigate various risks; and arbitrageurs seek to take advantage of mispricing or anomalies observed across various markets. Brokers often act as mediators that bring buyers and sellers together, earning a commission or fee for their services. They are third parties that facilitate trades between buyers and sellers but who do not take an actual position in a stock.

In a stock market, retail and institutional investors, traders, as well as market makers (MMs) and specialists take part and maintain liquidity and provide two-sided markets.

There are two important participants, first, those who have excess funds (the savers), and those who need funds (the borrowers). Bringing these two parties together are financial institutions and financial markets. The participants in the financial system deal in exchange of money, capital, assets, and securities. Financial institutions work in tandem with each other to create a fully interdependent financial system. The importance of financial institutions lies in the fact that they provide services which lead to a smooth functioning of the financial system. The financial institutions play the following role in the economic development of the country:

- 1) Help to form financial resources through mobilization of savings of the households, corporate and the public sector.
- 2) Channelise resources or investments to economic agents in different sectors of an economy: agriculture, manufacturing and services for cultivation of land, production of goods and provision of services.
- 3) Meet short and long-term financial needs.
- 4) Banking institutions multiply the monetary resources by the process of credit creation.
- 5) Transfer surplus funds from one part of the economy to another keeping in mind the national priorities and the market demand.
- 6) Encourage people to divert their physical assets into financial assets and make that available for growth of production, trade and commerce.
- 7) Provide a variety of financial assets to suit the different needs of investing public and corporate.
- 8) Facilitates mechanisms to control the risks and address uncertainties for the stability of the financial system. They reduce the risk of capital loss through diversification of funds to different industries.
- 9) Encourage entrepreneurial skills and assist with new projects.
- 10) Help accelerate the process of economic growth of the country.
- 11) Enable better decision-making.
- 12) Make financial resources available to the Government.

In this context, it is worthwhile to remember the recession of 2008-09 which underscores the importance of financial institutions to the economy. Businesses, for instance, depend on financial institutions for money. When they can't get it, production does not expand or may even be curtailed, unemployment rises, mortgage and other credit defaults increase, and people and businesses stop spending money, which reduces income for other people and businesses, and reduces tax revenue for governments, which causes them to cut spending, which causes more unemployment, etc. This is why governments around the world injected trillions of money into their financial

institutions during the recession to prevent their collapse and the subsequent stalling of the economy. Indeed, financial system with its institutions is the plumbing system of the economy and when it freezes or gets blocked the flow of money in the economy gets affected, undermining economic activity. It is desirable therefore to take all such measures that ensure the stability of the financial system in an economy to take proper measures.

Financial institutions serve most people in some way, as financial operations are a critical part of any economy, with individuals and companies relying on financial institutions for transactions and investing. They develop the financial securities and provide the financial markets where lenders, borrowers, investors, speculators, and hedgers can exchange money for future payments in the form of interest, for ownership interests, such as stocks, for the payment of future contingent claims, such as with options and derivatives, and for sharing risk, such as the pooling of insurance premiums for financial protection. Thus, financial institutions allow the exchange of funds between lenders, investors, and borrowers, the more efficiently this works the better it is for an economy.

1.4 FINANCIAL MARKETS

As discussed earlier, financial market is referred to a space, where the selling and buying of financial assets and securities take place. As noted, the trade of buying and selling *securities, derivatives, commodities, forex*, etc. is carried on in financial markets. Banks, financial institutions, insurance companies, mutual funds, pension trusts etc. are participants in financial markets.

Financial markets (bonds and stocks), instruments (derivatives, bank CDs, and futures), and institutions (banks, pension funds, insurance companies, and mutual funds) give investors the opportunities to choose specific services that meet their needs. Financial markets and financial institutions together contribute to economic growth. Financial markets are vital for the smooth operation of modern economies.

Financial market is a term used to denote the total infrastructure which facilitates the *trade of financial securities*. Examples of financial securities are *currency, bonds, stocks, derivatives, commodities, forex etc.* Businesses and investors can go to financial markets to raise money to grow their businesses and to make more money, respectively. They rely heavily on informational transparency to ensure that the markets set prices that are efficient and appropriate. The market prices of securities may not be indicative of their intrinsic value because of macroeconomic forces like the monetary and fiscal policies of the government.

The financial markets are broad in scope and scale. Despite covering many different asset classes and having various structures and regulations, all financial markets work essentially by bringing together buyers and sellers in

some asset or contract and allowing them to trade with one another. This is often done through an auction or price discovery mechanism.

The important participants in financial markets are intermediaries like banks, brokers, custodians, depositories, etc. Without intermediaries, the financial market cannot run seamlessly and efficiently. They are agencies which work in an assisting role. They execute the function of fund collection from lenders, and *disbursement of capital* to borrowers. They also *handle* financial securities like stocks, mutual fund units, bonds, deposit certificates etc., on behalf of the lender and borrowers.

Equilibrium in financial markets is usually determined by assuming that there is perfect competition, and by using the well-known tools of demand and supply. Equilibrium in financial markets is established when the expected demand for funds (credit) for short-term and long-term investments matches with the planned supply of funds generated out of saving and credit creation. In other words, the equality of total desired borrowing with the total desired lending is necessary for establishing an equilibrium rate of interest.

1.4.1 Types of Financial Markets

There are different types of financial markets. They are discussed as follows:

- **Over the Counter (OTC) Market:** They manage public stock exchange dealing usually with small companies that can be traded cheaply and have less regulation.
- **Bond Market:** A bond market is a place where investors loan money on bonds as a security for a set of times at a predefined rate of interest. Bonds are issued by governments, corporations, states, municipalities, etc.
- **Money Market:** They trade in products with highly liquid short-term maturities (of less than one year) and are characterized by a high degree of safety and a relatively low return on interest. At the wholesale level, the money markets involve low-volume trades between institutions and traders. At the retail level, they include money market mutual funds bought by individual investors and money market accounts opened by bank customers. Individuals may also invest in the money markets by buying short-term papers.
- **Derivatives Market:** A derivative is a contract between two or more parties whose value is based on an agreed-upon underlying financial asset (like a security) or set of assets (like an index). Derivatives are secondary securities whose value is solely derived from the value of the primary security that they are linked to. In and of itself a derivative is worthless. Rather than trading stocks directly, a derivatives market trades in *futures and options contracts*, and other advanced financial products,

that derive their value from underlying instruments like bonds, commodities, currencies, interest rates, market indexes, and stocks.

Future markets are where future contracts are listed and traded. Unlike forwards, which trade OTC, futures markets utilize standardized contract specifications, are well-regulated, and utilize clearing houses to settle and confirm trades. Options markets list and regulate options contracts. Both futures and options exchanges may list contracts on various asset classes, such as equities, fixed-income securities, commodities, and so on.

- **Commodities Markets** are venues where producers and consumers meet to exchange physical commodities such as agricultural products (e.g., corn, livestock, soybeans), energy products (oil, gas, carbon credits), precious metals (gold, silver, platinum), or “soft commodities” (such as cotton, coffee, and sugar). The bulk of trading in these commodities takes place in spot markets, where physical goods are exchanged for money. Forwards, futures, and options on commodities are exchanged both OTC and on listed exchanges around the world.
- **The Forex Market** is the market in which participants can buy, sell, hedge, and speculate on the exchange rates between currency pairs. The forex market is the most liquid in the world, as cash is the most liquid of assets. The forex market is also decentralized and consists of a global network of computers and brokers from around the world. The forex market is made up of banks, commercial companies, central banks, investment management firms, hedge funds, and retail forex brokers and investors.
- **Cryptocurrency Markets:** Since 2009, the world has seen the introduction and rise of cryptocurrencies such as Bitcoin and Ethereum. They are decentralized digital assets that are based on blockchain technology. Hundreds of cryptocurrency tokens are available and traded globally across a patchwork of independent online crypto exchanges. These exchanges host digital wallets for traders to swap one cryptocurrency for another, or fiat monies such as dollars or euros. Because the majority of crypto exchanges are centralised platforms, users are susceptible to hacks or fraud. Decentralized exchanges are also available that operate without any central authority. These exchanges allow direct peer-to-peer (P2P) trading of digital currencies without the need for an actual exchange authority to facilitate the transactions. Futures and options trading are also available on major cryptocurrencies.

Having this backdrop of types of financial markets, they can be further classified based on their nature of transactions:

1) Nature of Claim:

- a) Debt Market where fixed bonds and debentures are exchanged between investors.

- b) Equity Market is a place for investors to deal with equity.

2) Maturity of Claim:

- a) Money Market deals with short-term funds such as a certificate of deposits, treasury bills, and commercial papers, etc. which mature within twelve months.
- b) Capital Market trades medium and long-term financial assets, that is, securities having a maturity period of one year or more.

3) Timing of Delivery:

- a) Cash Market is a marketplace where trade is completed in real-time.
- b) Futures Market where the delivery or compensation of products is taken in the future specified date.

4) Organizational Structure:

- a) The exchange-traded Market has a centralised system with a patterned procedure.
- b) Over-the-Counter Market has a decentralised organization with customised procedures.

5) Nature of transaction

- a) The primary markets deal in new financial claims or new securities, and, therefore, they are also known as 'new issue markets'. They mobilize savings and they supply fresh or additional capital to business units.
- b) Secondary markets deal in securities already issued or existing or outstanding. They do not contribute directly to the supply of additional capital; they do so indirectly by rendering securities issued on the primary market liquid.

6) Regulation status

- a) Unorganized financial market is old Indigenous market mainly made of indigenous bankers, money lenders etc.
- b) Organized financial market comes under the regulatory purview of RBI and SEBI.

1.4.2 Importance and Functions of Financial Markets

The financial market allocates limited resources to the nation's economy. Without them, capital cannot be allocated efficiently. Economic activity such as commerce and trade, investment, and production would be greatly diminished. The financial market enables efficient trade of securities, and transfer of funds, between lenders and borrowers. It is the financial market which creates securities for investment. People who have surplus funds invest

in these securities to earn a return on their investments. Security markets are important for raising money for corporations and institutions and also for investors to allocate their money.

The financial market makes the transaction between savers and investors easy. It becomes a one-stop shop for all lenders/investors and borrowers. Hence the market becomes an effective source for capital generation and investment.

In short, there are many things that financial markets make possible. They provide a place where participants like investors and debtors, regardless of their size, will receive fair and proper treatment. They facilitate individuals, companies, and government organizations with access to capital. The financial markets help lower the unemployment rate because of the many jobs they offer. By allowing a free market for the flow of capital, financial obligations, and money the financial markets make the global economy run more smoothly while also allowing investors to participate in capital gains over time.

Financial markets exist for several reasons, but the most fundamental function is to allow for the efficient allocation of capital and assets in an economy. Mentioned below are the important functions of the financial markets:

- i) Create securities products that provide a return for those who have excess funds (Investors/lenders) and make these funds available to those who need funds (borrowers).
- ii) Mobilizes savings and trading it in the most productive methods.
- iii) Assist in price discovery of securities through interaction with the investors and depending on the demand and supply in the market.
- iv) Provide liquidity to bartered assets.
- v) Less time-consuming and cost-effective as parties don't have to spend extra time and money to find potential clients to deal with securities. They also decrease costs by giving valuable information about the securities traded in the financial market.

To recapitulate, financial markets put savings into more productive use, determine the price of securities, make financial assets liquid, and lower the cost of transactions.

Check Your Progress 1

- 1) What is the significance of the financial system?

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2) What is the financial market?

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3) What are the different types of financial markets?

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1.5 FINANCIAL INSTRUMENTS

Any asset which holds capital and can be traded in the market is referred to as a financial instrument. A financial instrument, an intangible asset is a claim against a person or an institution for the payment at a future date of a sum of money and/or a periodic payment in the form of interest or dividend. It is a tradable asset representing a legal agreement or a contractual right to evidence the monetary value/ownership interest of an entity. Financial instruments are typically traded in **financial markets** where the price of a security is arrived at based on the play of market forces.

Key financial instruments that are traded in the stock market include shares, stocks, bonds, derivatives, mutual funds, cheques etc. Investing in financial instruments carries a financial risk.

1.5.1 Types of Financial Instruments

There are typically three types of financial instruments: cash instruments, derivative instruments, and foreign exchange instruments. Furthermore, the financial instruments can be classified based on the 'asset classes' into 'equity-based' or 'debt-based'. If the instrument is debt, then it can be further categorized as 'short-term debt' and 'long-term debt'. Foreign exchange instruments and transactions are neither debt- nor equity-based and belong in their category.

Financial instruments can also be split into two categories: *Complex and Non-complex*. Complex financial instruments require in-depth knowledge for traders to be successful when trading them. The most commonly traded complex financial instruments are derivatives. These can be contracts for differences (CFDs), futures contracts and options. Different derivatives have different benefits. Non-complex financial instruments can be traded without a

vast amount of specialist knowledge which include shares or equity securities, as well as debt securities and certain types of investment funds.

Further, financial securities may be primary or secondary securities. Primary securities are also termed direct securities as they are directly issued by the ultimate borrowers of funds to the ultimate user. Secondary security is a term used to explain the financial market where securities are purchased from other investors.

Cash instruments: Cash instruments are those whose value is determined directly by the markets. They can be securities, which are readily transferable and loans and deposits, where both borrower and lender have to agree on a transfer.

Derivatives: A derivative is a financial instrument such as options, forward exchange contracts etc. issued for various types of underlying assets, whose value depends on or is derived from one (or more) of underlying instruments.

Check Your Progress 2

- 1) Define financial institutions and also differentiate between different types of financial institutions.

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- 2) Define financial instruments. Distinguish between primary securities and secondary securities.

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- 3) What are financial services? Give some examples.

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1.6 LET US SUM UP

In this unit, we discussed the nature of the financial system. It is a complex, well-integrated set of sub-systems of financial institutions, markets, instruments, and services which facilitate the transfer and allocation of funds,

efficiently and effectively. Two important classifications of financial institutions are Banking institutions and non-banking institutions and Intermediaries and non-intermediaries. The units outlined financial markets as mechanisms enabling participants to deal with financial claims. Two important classifications of financial markets are Primary market and secondary market, Money market and Capital market. The unit also deliberated about the financial instrument as a claim against a person or an institution for the payment at a future date of a sum of money and/or a periodic payment in the form of interest or dividend. We discussed financial services rendered by financial intermediaries that bridge the gap between the lack of knowledge and information on the part of investors in the face of growing sophistication of financial instruments and markets.

1.7 KEY WORDS

Cryptocurrency	: Decentralized digital assets that are based on blockchain technology.
Derivative	: It is a financial instrument such as options, forward exchange contracts etc. are issued for various types of underlying assets, whose value depends on, or is derived from one (or more) of underlying instruments.
Financial institutions	: All kinds of organizations intermediate and facilitate financial transactions of both individual and corporate customers.
Financial instrument	: is a claim against a person or an institution for the payment at a future date of a sum of money and/or a periodic payment in the form of interest or dividend.
Financial market	: It is referred to as space, where the selling and buying of financial assets and securities take place. The trade of buying and selling securities, derivatives, commodities, forex, etc. is carried on in financial markets.
Financial system	: It is a complex, well-integrated set of sub-systems of financial institutions, markets, instruments, and services which facilitate the transfer and allocation of funds, efficiently and effectively.
Primary markets	: deal in the new financial claims or new securities, and, therefore, they are also known as ‘new issue markets’.
Primary securities	: They are direct securities as they are directly issued by the ultimate borrowers of funds to the ultimate user.

- Secondary markets** : deals in securities already issued or existing outstanding.
- Secondary security** : is a term used to explain a financial market where securities are purchased from other investors.

1.8 SOME USEFUL REFERENCES

Houthakker, H. S., & Williamson, P. J. (1996). *The Economics of Financial Markets*. Oxford University Press, USA.

Mishkin, F. S., & Eakins, S. G. (2006). *Financial Markets and Institutions*. Pearson Education India.

1.9 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) A financial system is a complex, well-integrated set of sub-systems of financial institutions, markets, instruments, and services which facilitate the transfer and allocation of funds, efficiently and effectively between different economic agents.
- 2) Mechanism enabling participants to deal with financial claims.
- 3) A. Primary market and secondary market and B. Money market and Capital market.

Check Your Progress 2

- 1) Financial institutions are business organisations that act as mobilisers and depositors of savings and as purveyors of credit or finance. Banking Institutions are “creators” of credit, and the non-banking Institutions as “purveyors” of credit. Intermediaries intermediate between savers and investors while non-intermediary institutions are in the lending business.
- 2) A claim against a person or an institution for the payment at a future date of a sum of money and/or a periodic payment in the form of interest or dividend. Primary securities are also termed direct securities as they are directly issued by the ultimate borrowers of funds to the ultimate user. Secondary security is a term used to explain the financial market where securities are purchased from other investors.
- 3) facilitating transactions, mobilizing savings, allocating capital, monitoring management.

UNIT 2 FINANCIAL INTERMEDIATION

Structure

- 2.0 Objectives
- 2.1 Introduction
- 2.2 Concept of Financial Intermediation
 - 2.2.1 Direct and Indirect Finance
 - 2.2.2 Intermediaries and Non-intermediaries
- 2.3 Types of Financial Intermediaries
- 2.4 Function and Roles of Financial Intermediaries
- 2.5 An Overview of the Indian Financial System
- 2.6 Regulation of Financial Institutions
- 2.7 Let Us Sum Up
- 2.8 Suggested Readings
- 2.9 Key Words
- 2.10 Answers/Hints to Check Your Progress Exercises

2.0 OBJECTIVES

After going through this unit, you will be able to:

- Explain the concept of direct and indirect finance;
- Examine the concept of financial intermediaries;
- Describe the features of financial intermediaries; and
- Provide an overview of the evolution of the Indian Financial System.

2.1 INTRODUCTION

As discussed in Unit 1, the financial system consists of financial institutions, financial markets, financial services, financial instruments, financial practices and financial transactions. We already know that the term financial institutions refer to all kinds of organizations which intermediate and facilitate financial transactions of both individual, corporate and public entities. Many financial institutions perform the role of financial intermediaries. They are intermediaries between the savers and borrowers of funds. Financial institutions have several functions that assist the economic agents in fulfilling their various needs. They deal in all types of finance-related businesses and gaining prominence in broadening the finance-related services in the country. They play a crucial role in the

economy. In this unit, we will study more about financial intermediaries and provide an overview of the Indian financial system.

2.2 CONCEPT OF FINANCIAL INTERMEDIATION

2.2.1 Direct and Indirect Finance

The two primary ways through which funds are channelled from savers to borrowers in an economy are direct and indirect finance. These concepts relate to the mechanisms by which financial resources move between those who have excess funds (savers) and those who need funds (borrowers). In direct finance, funds are directly provided by savers to borrowers in the financial markets without the involvement of any intermediary. In indirect finance, there is a special kind of 'middleman', a third party that stands between borrowers and lenders. This 'middleman' is known as a financial intermediary. The role of a financial intermediary is to accumulate funds from parties that have excess funds (the savers) and make those funds available to those who have deficient funds and need those funds (the borrowers). While we will discuss various types of financial intermediaries shortly, let us discuss here one of the financial intermediaries: banks. Banks are financial intermediaries because they accept deposits, and then provide loans from these funds to borrowers. Banks pay depositors interest on their deposits and receive interest payments from borrowers on the loan advanced. The bank is in the middle, as an intermediary and aims to earn a profit on the spread between the rate of interest charged to borrowers and the rate of interest paid to depositors.

As mentioned, a financial intermediary is like a middleman; so is a broker who facilitates exchange between borrowers and lenders in financial markets where direct finance is involved. But there is a difference between brokers and financial intermediaries: a broker brings together borrowers and lenders but does not lend to borrowers or issue bonds. All that a broker does is facilitate transactions involving direct finance and charge a commission for his services rendered. A broker does not create any financial instrument. In contrast to this, financial intermediaries lend to borrowers by creating and issuing their securities, for example: bonds. A financial intermediary does not merely bring borrowers and lenders together; financial intermediaries themselves give loans to lenders. They earn profits by repackaging deposits of savers into loans, and themselves lending to borrowers.

2.2.2 Intermediaries and Non-intermediaries

Intermediaries intermediate between savers and investors; they lend money as well as mobilize savings; their liabilities are towards the ultimate savers, while their assets are from the investors or borrowers. Non-intermediary institutions do the loan business but their resources are not directly obtained from the savers. All banking institutions are intermediaries, many non-banking institutions also act as intermediaries and when they do so, they are known as **non-banking financial intermediaries**.

Financial intermediaries provide key financial services such as merchant banking, leasing, hire purchase, credit-rating, etc. Financial services rendered by the financial intermediaries' bridge the gap between the lack of knowledge and information on the part of investors and growing sophistication of financial instruments and markets.

It may be recalled from Unit 1, financial intermediaries-banking and non-banking — as financial institutions, perform the following functions:

- i) They act as a bridge between savers and investors.
- ii) They reduce the cost of financial services provided.
- iii) They play an important role in the development of financial services in the country.
- iv) They provide consultancy services to clients on their investments related to the financial markets where a huge amount of risk is involved. In this way, they improvise the decision-making process.
- v) They profit by earning higher returns on their investments than they pay for their sources of money.
- vi) The assets of a financial intermediary are the loans, stocks, bonds, and real estate that are the company's investments and its liabilities are its obligations to its customers, which include deposits, insurance policies, and pension payouts.

Check Your Progress 1

- 1) State the meaning of finance and financing.

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- 2) What are intermediaries and non-intermediaries?

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- 3) Differentiate between direct and indirect finance.

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2.3 TYPES OF FINANCIAL INTERMEDIARIES

As we discussed earlier, financial intermediaries play a very important role in the financial system by facilitating the flow of funds between savers and

borrowers. There are various types of financial intermediaries, each serving a specific function in the financial markets. Some of them are:

- Commercial Banks; e.g. State Bank of India, HDFC Bank, ABN AMRO Bank, PNB, HSBC Bank, etc. They are traditional financial intermediary that accept deposits from savers and provide loans to individuals, businesses, and governments.
- Development Banks, e.g. IDBI, NABARD, SIDBI, etc.
- Finance companies
- Insurance Companies collect premiums from policyholders in exchange for coverage against various risks and invest the collected premiums to generate returns.
- Life Insurance Companies; LIC, Allianz, AIG etc.
- Mutual Funds, Standard Chartered Mutual Fund, Morgan Stanley Mutual Fund, SBI Mutual Fund etc.
- Pension Funds; e.g. EPF (Employees Provident Fund), PPF (Public Provident Fund). They manage retirement savings on behalf of individuals and employees. They invest these funds in a diversified portfolio of assets, including stocks, bonds, and real estate, to generate returns that can be used to pay retirees.
- Credit Unions. They are similar to banks and offer a range of banking services. However, they are typically owned by their members as cooperative societies and operate as nonprofit organizations.

2.4 FUNCTIONS AND ROLES OF FINANCIAL INTERMEDIARIES

Financial institutions as intermediaries work in tandem with each other to create a fully interdependent financial system. The importance of financial institutions lies in the fact that they provide services which lead to a smooth functioning of the financial system. They allow the exchange of funds between lenders, investors, and borrowers. There are three aspects to the process by which saving is channelled to investors through the financial system: (i) the amount of savings and the resulting investment that are raised in the economy, (ii) the question of who saves and who invests and how do they do that respectively, and finally, (iii) the flow of funds through financial institutions and markets as money passes on its way from savers to investors. Financial institutions have specific function in all these stages in an economy and how well the financial system and its institutions are organised to facilitate these stages contributes to the growth of economic activity. Some of these functions are as under:

Directing the Payment System: One of the primary functions of financial institutions is managing the payment system. Payment system refers to

everyday commercial transactions that involve individuals and businesses. Financial institutions keep the payment system in motion through checking and savings accounts, debit and credit cards, and wire transfers. These methods of exchange allow individuals, businesses and institutions to handle their financial transactions daily.

Assisting with Resources and Capital: Financial institutions help individuals and corporations with resources and capital management by extending credit to those who can pay it back. Banks and other institutions can pool resources together to allow others to borrow money. Loans and credit cards allow families and companies to borrow funds and pay them back on a regular schedule. Acquiring capital for a new or existing business or personal project can be difficult, so financial institutions help people and businesses have access to the capital they need to be successful.

Moving Financial Resources: Financial institutions provide the movement of resources from place to place. These institutions assist with larger transfers of funds like corporate investments, purchases of real estate, and construction loans, as well as other larger transactions. Financial institutions can transfer resources from one party to another more easily and with more flexibility than individuals or corporations can, which makes this function so crucial.

Risk Management: Financial institutions handle risk and uncertainty for businesses and households. Insurance companies and other entities in the financial sector allow many individuals to collectively pool and distribute risk, making it easier to manage accidents and various difficulties that occur in professional or personal contexts.

Informing Financial Decisions: Financial institutions provide key information that makes decisions easier. Whenever key monetary decisions are made, one can very well appreciate the importance of such information. The information that financial institutions provide allows individuals and corporations to make informed and secure decisions in real time. They give a high-rated consultancy to the customers for their beneficial investments.

Sustaining the Market: Financial institutions also make it possible for individuals and other entities to invest in the stock market. Investors can make long-term or short-term ventures into the market for the promise of a higher return. Institutions like brokers and stock exchanges allow companies to issue stock to have more cash flow based on investors purchasing stock. Often, the stock market drives the pulse of the financial sector as a whole.

These days' financial institutions have a very broad spectrum of functions and with creativity and innovation, new areas of activities are being added regularly. They serve as a depository for their customers and make efforts to minimize the monitoring cost for customers.

In short, all the finance-related work such as managing assets, rate of return, liquidity, mobilization of savings, capital formation, maturity and risk

transformation, lowering the cost of a transaction, smoothening payment mechanism, etc. are done by the financial institutions for their customers.

2.5 AN OVERVIEW OF INDIAN FINANCIAL SYSTEM

The Indian financial system exhibits two prominent segments: the organized sector and the unorganized sector. The organized sector includes commercial, development, and cooperative banks, the stock market, and diverse non-banking financial institutions like insurance corporations and mutual funds. The unorganized sector is the informal credit market which provides loans to farmers, small enterprises and individuals. These two sectors vary majorly in terms of rates of interest and credit conditions.

The financial system was underdeveloped until independence. After 1947, the government started creating new financial institutions for the supply of finance for both industrial and agricultural purposes. For this purpose, several steps have been taken since independence. Some of them are:

- 1) ***Nationalization of financial institutions:*** Some important financial institutions at that time were nationalized. For example,
 - a) In the year 1948, the Reserve Bank of India (which was established in the year 1935 as a private-sector central bank) was nationalized.
 - b) In the year 1955, the then Imperial Bank of India was nationalized and renamed as State Bank of India.
 - c) In the year 1956, 245 life insurance business entities (consisting of 154 life insurance companies, 16 foreign companies and 75 provident companies) were nationalized and merged to form Life Insurance Corporation(LIC) of India.
 - d) In the year 1969, 14 Commercial Banks were nationalized.
 - e) In the year 1972, 107 general insurance business entities (consisting of 55 Indian insurance companies and 52 other general insurance operations of other companies) were nationalized and merged to form the General Insurance Corporation (GIC) of India.
 - f) In the year 1980, another 6 Commercial Banks were nationalized.
- 2) ***Establishment of new institutions:***
 - i) *Unit Trust of India (UTI)* – UTI was established in the year 1964 to strengthen the Indian financial system and supply institutional credit to industries. It was entrusted with the work of mobilisation of the savings of the public and provision of credit facilities to institutions for productive purposes. It has established the following subsidiaries:

- a) The UTI Bank Ltd.,
 - b) The UTI Investor Service Ltd.,
 - c) The UTI Security Exchange Ltd.,
- ii) *Development Banks* – Development banks are the institutions that provide medium and long-term finance for agriculture and industrial development purposes. These are established to provide not only credit facilities but also assist in discovering investment projects, preparing project reports, arranging technical advice, managing industrial units, etc. They are also intended to develop backward regions as well as small and new entrepreneurs. The development banks that were established over the years are as under:
- a) The Industrial Finance Corporation of India (IFCI) at the central level was established in the year 1948 to provide medium and long-term finance to industrial concerns.
 - b) The State Financial Corporations (SFCs) at the state level were established under the State Financial Corporation Act, of 1951 to provide medium and long-term finance to medium and small industries in the respective states. The State Industrial Development Corporation / State Industrial Investment Corporation (SIIC) were also set up.
 - c) The Industrial Credit and Investment Corporation of India (ICICI) was established in the year 1955 to develop large and medium industries in the private sector. (Now this is merged with ICICI Bank and it is a private commercial bank now).
 - d) The Refinance Corporation of India (RCI) was established in the year 1958 to provide refinance facilities to banks against term loans granted by them to medium and small units. (Now this is merged with the Industrial Development Bank of India (IDBI).
 - e) IDBI was established in the year 1964 as a wholly owned subsidiary of the Reserve Bank of India to act as the apex institution in the area of development banking and coordinate the activities of all the other financial institutions. (Now the ownership of IDBI is with the central government).
 - f) Small Industries Development Bank of India (SIDBI) established in the year 1990 under the Act, is the principal financial institution for the promotion, financing and development of the micro, small and medium enterprises (MSMEs) sector and for the co-ordination of the functions of the institutions engaged in similar activities.

- g) *Agriculture Financing Institutions*: The Agricultural Refinance and Development Corporation (ARDC) was established in the year 1963 to finance major development projects like minor irrigation, farm mechanization, land development, horticulture, dairy development, etc. However, the National Bank for Agriculture and Rural Development (NABARD) was established in the year 1982 and the ARDC was merged with it. Now the whole sphere of agricultural finance has been entrusted with NABARD.
- h) *Export and Import Bank of India (EXIM Bank)*: EXIM Bank was established in the year 1982 to take over the operations of the International Finance Wing of IDBI. Its primary objective is to provide finance to exporters and importers. It also functions as the principal financial institution for coordinating the working of other institutions engaged in the financing of foreign trade.
- i) *National Housing Bank (NHB)*: The National Housing Bank was established in the year 1988 as an apex institution to mobilize resources for the housing sector and to promote housing finance institutions both at regional and local levels. It provides refinancing, underwriting and guaranteeing facilities to housing finance institutions and coordinates the working of all agencies connected with housing finance.
- j) *Stock Holding Corporation of India Ltd. (SHCIL)*: SHCIL was established in the year 1987 to tone up the stock and capital markets in India. It provides quick share transfer facilities, clearing services, depository services, management information services and development services to both individual and corporate investors.
- k) *Encouragement of financial services and new instruments*:
 - i) *Mutual Funds Industry*: Both private and public sector financial institutions are being encouraged to float mutual funds.
 - ii) *Venture Capital Industry*: Both private and public sector financial institutions are being encouraged to finance through venture capital.
 - iii) *Credit Rating Agencies*: Credit rating agencies like Credit Rating and Information Services of India Ltd., (CRISIL), Investment Information and Credit Rating Agency of India Ltd., (ICRA) and Credit Analysis and Research Ltd., (CARE) are established to help investors make the decisions of their investment and also to protect them from risky ventures.

- iv) *Introduction of new financial instruments* – New and different types of financial instruments like public sector bonds, national savings certificates, post office savings schemes, different varieties of shares and debentures, different schemes of insurance, and different types of bank deposits, are being introduced to cater to the needs of different investors.

The structure and evolution of the Indian financial system in the late 1960s closely reflected a growing degree of public ownership and control of financial institutions in keeping with the political priorities of the time. The extensive control of the financial system was aimed at ensuring the flow of finances in the desired directions and in particular to priority areas of development. With the economic reforms of 1991, the financial system has undergone a structural change with the private sector now playing a substantial role. However, it suffers from one major weakness i.e., high over dues or non-performing assets of the lenders which affects many Indian financial institutions. It is a problem in both industrial and agricultural finance although the situation is on the mend in the recent years.

In short, the evolution of the financial system since 1950 has increased the “financialization” of the Indian economy. From a gross saving rate of about 8 per cent for the country at the time of independence, it has gone up to 35 per cent at its highest, declining to about 30 per cent in the most recent years. Importantly, from a preponderance of physical saving at over 90 per cent in the gross saving rate of the country, at the time of independence, it has come down to about 50 per cent in the recent years. It indicates that an increased role of the public sector in the financial system has not retarded its development. Underpinning this financial deepening is an increase in the assets of the commercial banks. There is an increase in the importance of financial institutions in the economy and of financial flows relative to economic activity. This indicates that the financial system is playing an increasingly active role in the transfer of funds from the surplus to the deficit sectors in the economy. Financial intermediation in India’s *organized* sector is conducted by a wide range and number of financial institutions which have largely operated in major cities and towns but with growing presence in rural areas, aided by the expansion of digital infrastructure across the country, in recent years.

Check Your Progress 2

- 1) What are the important functions of financial institutions?

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- 2) Explain the significance of the financial system.

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- 3) Briefly provide an overview of India's financial system.

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2.6 REGULATION OF FINANCIAL INSTITUTIONS

The government needs to oversee and regulate banks and financial institutions because they play a critical role in the economy. Historically, bankruptcies of financial institutions have created panic and instability in the financial system. A well-regulated financial system inspires confidence among investors, depositors, and the general public. The regulation of financial institutions in India is overseen by several regulatory bodies. These organisations play a crucial role in streamlining the functioning of the financial system and protecting the interests of customers. The regulation of financial institutions in India is important for various reasons. Regulation helps maintain the stability of the financial system by preventing excessive risk-taking, fraud, and other activities that could lead to systemic crises. Regulations in the financial sector also aim to safeguard the interests of consumers especially small savers. The interest of the customers can be protected by operating transparently and adhering to ethical business practices.

Financial institutions are subject to regulations that require them to implement robust risk management practices. This helps prevent excessive risk-taking and ensures that institutions are adequately prepared for adverse economic conditions.

Another important reason for financial institutions being highly regulated is that most people will not be confident in investing in an unregulated institution. Thus, the regulation of financial institutions also aims to gain and retain customer confidence. Confidence in turn is crucial for the smooth functioning of financial markets and institutions.

In every country, government agencies oversee the regulation of financial institutions, formulating rules and guidelines for the sector while also ensuring compliance through monitoring.

In India, some key regulatory bodies are:

- Reserve Bank of India (RBI) formulates and implements monetary policy, issues currency, regulates the money supply and supervises banks and other financial institutions.
- Securities and Exchange Board of India (SEBI) regulates activities such as the issuance and trading of securities, as well as the functioning of mutual funds and portfolio managers.
- The Insurance Regulatory and Development Authority of India (IRDAI) regulates and oversees the insurance industry in India.
- PFRDA: Pension Fund Regulatory and Development Authority 2003.
- SIDBI: Small Industries Development Bank of India 1990; National Bank for Agriculture and Rural Development (NABARD): NABARD regulates and supervises cooperative banks and regional rural banks (RRBs) to ensure the flow of credit to rural areas for agricultural and rural development.
- NHB: National Housing Bank, 1988 to promote housing finance (RBI).
- Insolvency and Bankruptcy Board of India set up 2016: regulator for overseeing insolvency proceedings and entities like Insolvency Professional Agencies, Insolvency Professionals and Information Utilities in India.

Check Your Progress 3

1) Why is there a need for regulation of financial institutions?

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2) In what ways financial institutions are being regulated?

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3) Write a note on some of the key financial regulatory bodies in India.

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2.7 LET US SUM UP

A Financial Institution (FI) is an entity engaged in the business of dealing with financial and monetary transactions such as deposits, loans, investments, and currency exchange. They encompass a broad range of business operations within the financial services sector including banks, trust companies, insurance companies, brokerage firms, and investment dealers. FIs can vary by size, scope, and geography. FIs offer a wide range of products and services for individual and commercial clients. The specific services offered vary widely between different types of financial institutions.

When each agent and constituent functions well, the financial system as a whole is more stable. The health of the financial sector rests on a delicate balance between the various functions of financial institutions. Problems with one function can reflect on the other capacities of the financial system as well.

The financial institutions provide the best instruments to save and invest money and to earn good returns from that investment. They try to help nations in building up economies. However, there could be some risks associated with the functioning of some institutions at times and when there is a default, in the system there could be distress and loss for economic agents. It is therefore important to have a robust regulation for the a stable functioning of a financial system. Indeed, the health of a nation's banking system is a linchpin of economic stability.

2.8 SUGGESTED READINGS

Houthakker, H. S., & Williamson, P. J. (1996). *The Economics of Financial Markets*. Oxford University Press, USA.

Mishkin, F. S., & Eakins, S. G. (2006). *Financial Markets and Institutions*. Pearson Education India.

2.9 KEY WORDS

Direct finance : Funds are directly provided by savers to borrowers in the financial markets without the involvement of any intermediary.

Indirect finance : Funds are provided by 'middleman', a third-party that stands between borrowers and lenders.

Intermediaries : They intermediate between savers and investors; they lend money as well as mobilize savings.

Financial Institution (FI) : A company engaged in the business of dealing with financial and monetary

transactions such as deposits, loans, investments, and currency exchange.

Financial Intermediation : The financial intermediation process channels funds between third parties with a surplus and those with a lack of funds.

2.10 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) Finance means Financial resources Financing involves intermediation i.e., providing finance.
- 2) Refer to Sub-Section 2.2.2
- 3) Indirect finance, funds are directly provided from savers to borrowers. Indirect finance involves intermediary.

Check Your Progress 2

- 1) Directing the payment systems, assisting with resources and Capital, Risk management.
- 2) Refer to Section 2.4
- 3) Refer to Section 2.5

Check Your Progress 3

- 1) Prevent excessive risk taking, gain customer confidence, ensure compliance.
- 2) Refer to Section 2.6
- 3) Refer to Section 2.6

UNIT 3 BASIC BUSINESS ACCOUNTING

Structure

- 3.0 Objectives
- 3.1 Introduction
- 3.2 Basic Concepts of Accounting
 - 3.2.1 The Balance Sheet
 - 3.2.1 The Cash Flow Statement
 - 3.2.1 The Income Statement
- 3.3 Real Assets and Financial Assets
- 3.4 The National Accounts
- 3.5 Flow of Funds Accounts
- 3.6 The Relationship between Stocks and Flows
- 3.7 Exchange Rates
- 3.8 Rate of Interest
- 3.9 Government Borrowings
- 3.10 Nominal and Real Interest Rates
- 3.11 Let Us Sum Up
- 3.12 Key Words
- 3.13 Some Useful References
- 3.14 Answers/ Hints to Check Your Progress Exercises

3.0 OBJECTIVES

The main objective of this Unit is to:

- understand the framework of accounting, focusing on its process and the concept of financial statements;
- recognise the importance of financial markets in the Indian economy through a focus on real assets and financial assets;
- interpret the macroeconomic analysis of flows which incorporates national accounts, flow of funds accounts, international transactions etc. and
- know the certain features of the rate of interest.

3.1 INTRODUCTION

The main objective of this unit is to illustrate the basics of accounting which is usually recognised as the language of business. In financial terms, it

implies that accounting is the process of recording and reporting of business transactions (monetary terms) which enable users (owners, management, employees, banks etc.) to make appropriate judgements and decisions. Accounting is a significant tool to understand the functioning of financial markets. The concept of stocks and flows play an important role in grasping the economic meaning of accounting.

The accounting process normally starts with an economic activity which is then originated in the form of a document or a voucher. Several copies of the voucher are sent to the accounts department where they are recorded, analysed and kept in the books of accounts. After the recording of the transaction, classification is the next step which includes posting transactions in ledgers and further summarising them in the form of trial balance, trading profit and loss account and balance sheet. The next step is the analysis and interpretation and finally communication of the essential information to the users for financial decision-making.

It is necessary to understand the difference between stocks and flows to understand accounting. Thus, these concepts are well defined in Section 3.2 illustrating the statements such as balance sheet, income statement and cash flow statement.

The purpose of Section 3.3 is to outline a framework of real assets and financial assets and their disclosure in the balance sheet. Real assets are the assets used to produce goods and services. Financial assets are considered as claims on real assets or the income generated by them. The building is a real asset that is created. On the other hand, the loan is a financial asset which is designed by the transaction.

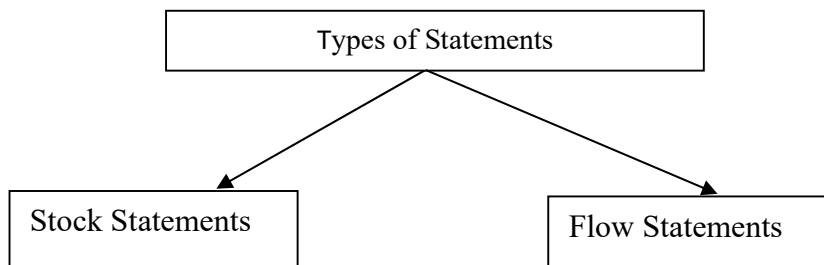
A macro-economic analysis is explained in Section 3.4 which covers national income accounts and national product accounts. Another tool of macro-economic analysis is the flow of funds accounts which is described in Section 3.5. Several indicators of financial development are also emphasized.

A fundamental distinction is made between stock and flow. Both concepts are used in calculating the national income of the country. For instance, saving is recognized as stock and investment is recognized as flow. The relationship between savings and investment is described in Section 3.6.

The importance of exchange rates in the international financial markets is explained in Section 3.7. The rate of interest is well-defined in Section 3.8 regarding the barter system and its impact on the economy. Rates of interest further emphasize how the emergence of the monetary system influences it. Government borrowing represents the government's process of raising money by issuing bonds or other securities to investors and it discusses in Section 3.9 that how government borrowings are financed. Lastly, the importance of nominal and real interest rates in the financial system is being elaborated in Section 3.10.

3.2 BASIC CONCEPTS OF ACCOUNTING

Accounting concepts imply the basic assumptions which are the building blocks of any business entity. Based on these assumptions, financial statements are prepared. In the context of accounting, two types of statements are reflected in the figure



The balance sheet is usually referred to as the stock statement. Flow Statements are further demarcated into two kinds i.e., cash flow statement and income statement. These statements are meant for business firms as well as for individuals.

3.2.1 The Balance Sheet

Balance Sheet is the statement which discloses the financial condition of the firm at a specific point in time. The balance sheet aims to ascertain the amount of assets the firm owns and the amount of liabilities it owes to outside parties. Assets are the resources owned by the business expecting to have benefits in the future. On the other hand, liabilities are represented as the borrowings or the obligations of the firm. Net worth is the difference between assets and liabilities which reflects the value of the firm. An illustrative balance sheet of firm is depicted in the table 3.1.

Table 3:1 Balance Sheet as on 2020 and 2021 (millions)

Particulars	2020	2021
Assets		
Cash	70	90
Accounts Receivable	210	200
Land and Building	180	180
Plant and Equipment	700	720
Inventories	160	140
Total Assets	1320	1330
Liabilities		
Loan from Bank	400	300
Accounts Payable	100	150

Mortgages and Bonds	560	540
Total Liabilities	1060	990
Net Worth	260	340

Cash is an important component for every firm which includes currency and readily available bank deposits. Accounts payable implies creditors and bills payables i.e., payable to the firm's suppliers. Bank loans also represent a liability. Accounts receivables imply debtors and bills receivables i.e., the amount received from the firm's customers. Some adjustments might be required for bad debts (amount uncollected from debtors or customers). Taking into consideration the value of the firm, land and building as well as plant and equipment are reflected at their historical costs as per the accounting convention. However, certain economists would value the land, plant and equipment at current value. Therefore, to avoid this complexity, the firm's financial statements are being prepared in conformity with Generally Accepted Accounting Principles (GAAP). The purpose of these principles is to prohibit misrepresentation and to provide accounting insight. This is also shown in the table that the assets such as plant and equipment and other categories normally lose their value because of wear and tear which is called depreciation. Depreciation is normally deducted from the value of the asset and the balance sheet reflects the depreciated value of the asset. Inventories which are also in the category of assets comprise raw materials (inputs), finished products (produced but not yet sold) and work in progress (in process).

Lastly, the category of liabilities is the mortgages and bonds which are shown as face or par value as per GAAP. However, bond prices normally fluctuate if traded in the credit or bond market. The firm would value the bonds at the market price if it is less than the face value of the bond as the bonds can be bought back at the price by the debtor.

3.2.2 The Cash Flow Statement

A cash flow statement is a statement of inflows and outflows of cash of the business firm during the accounting period. This statement summarises the cash transactions into certain categories such as operating, investing and financing activities.

Operating activities are involved in the activities which produce revenue such as cash receipts from the sale of goods and providing services, cash payments to employees etc.

Investing activities comprise of purchase and sale of long-term assets such as cash payments to purchase fixed assets.

Financing activities involve those activities which result in changes in the size and composition of the owner's capital (both preference and equity

shares) and borrowings of the firm such as cash received from issuing shares, payment of cash dividends etc.

3.2.3 The Income Statement

Though the cash flow statement provides information about the cash flows of the firm during a period this information is restricted to the extent of cash flows from operating, investing and financing activities. This statement neither reflects the revenues recognized nor the expenses incurred during the period. It is mainly the income statement which displays the revenues recognized and expenses incurred. Revenues are generally recognized in the period in which goods are delivered and services are offered. It is not necessary to make payment at that period. Expenses are used in the generation of revenues irrespective of whether the cash was used to pay those expenses.

The income statement is prepared by the business firm before the cash flow statement is prepared.

Therefore, there is no direct relationship between the income earned and cash flows associated with the period.

3.3 REAL ASSETS AND FINANCIAL ASSETS

Real assets and financial assets are two different kinds of assets, each with its own features and investment opportunities.

Real assets are tangible assets, such as real estate, land, machinery, and commodities, that have intrinsic (fair) value due to their physical properties or usefulness. These assets can generate income or appreciate over time. These assets are often used as a hedge against inflation. Financial assets, on the other hand, are intangible assets, such as stocks, bonds, mutual funds, and currencies, that represent a claim to expected cash flows or certain ownership rights. These assets have a major hold in the financial markets and demand and supply forces determine the value of financial assets.

The relationship between real assets and financial assets is complex and dynamic. In general, the performance of real assets can impact the performance of financial assets and vice versa. For example, a rise in the value of real estate appreciated the value of real estate investment trusts (REITs), which are financial assets that invest in real estate properties. Similarly, fluctuations in commodity prices can influence the performance of mutual funds which invest in commodities. Financial assets, such as stocks and bonds, can also impact real assets. For example, changes in interest rates can impact the demand for real estate, as higher interest rates can make it more expensive for people to borrow money to buy a house. Stock market fluctuations can also impact investor confidence, which can affect investment decisions in real assets.

To conclude, it can be said that both real assets and financial assets have different investment opportunities and bear their risks and returns. A balanced investment strategy can be achieved by incorporating a mix of both assets by diversifying into portfolios.

Disclosure of Real Assets and Financial Assets in the Balance Sheet of the Firm

Because of varying characteristics, both real assets and financial assets are disclosed separately in the balance sheet of the firm. Real assets, such as property, plant, and equipment (PPE), are typically known as long-term assets and are shown under the non-current assets of the balance sheet. PPE includes assets that have a useful life of more than one year, such as land, buildings, machinery, and equipment. These assets are usually reported at their historical cost less accumulated depreciation, which reflects the wear and tear or obsolescence of the assets over time. Real assets can also include other long-term assets such as patents, copyrights, and trademarks, which are intangible but still have a real value to the company.

Financial assets, such as cash, accounts receivable, and investments in securities, are typically classified as either current or non-current assets, depending on their liquidity and expected holding period. Current assets are assets that are expected to be converted into cash within one year, while non-current assets are assets that are expected to be held for more than one year. Financial assets are usually reported at their fair market value or market price, which reflects the current market value of the asset at the time of the balance sheet.

Check Your Progress 1

- 1) What is meant by accounting? Demonstrate the types of statements prepared by the business entity during an accounting period.

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- 2) Discuss the relationship between real assets and financial assets and how they are disclosed in the balance sheet of the firm.

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3.4 THE NATIONAL ACCOUNTS

The macroeconomic analysis is best known through the system of National Income Accounts. We have already learned concept related to National Income in macro economics let us briefly discuss them again. Production is measured in the economy and earnings are also measured that are derived from production.

The concept of flow is classified into two: National Product and National Income which are used to measure the overall economic performance of the country.

National income refers to the total income earned by a country's residents. This comprises wages, salaries, profits, and rental income. It is computed by summing up all the income earned in a country during a given period, usually a year. A country's economic well-being is measured by the national income which shows the capability of the residents to consume goods and services.

National product, on the other hand, is the total value of all goods and services produced within a country's borders in a given period, usually a year. It is computed by summing up the value of all final goods and services produced in a country, regardless of who produced them. National product comprehends both goods that are traded in the market and those that are meant for personal use, like homemade vegetables. National product is measured by Gross Domestic Product (GDP) and Gross National Product (GNP).

Gross Domestic Product (GDP) is a broad measure of the output of goods and services in an economy. It is generally the compilation of value-added output. GDP is measured either through the production side or through the consumption side. GDP is categorized into three sectors if measured through production i.e., agriculture, industry and services. However, from the consumption or expenditure point of view, GDP is the summation of private consumption, government consumption, investment and net exports.

- *Private Consumption:* This is the amount spent by households on goods and services, including durable goods (such as cars and appliances), non-durable goods (such as food and clothing), and services (such as healthcare and education).
- *Investment expenditure:* This is the amount spent by businesses on new capital goods, such as machinery and buildings, as well as on research and development.
- *Government expenditure:* This is the amount spent by the government on goods and services, including public infrastructure, defence, and social programs.
- *Net exports:* This is the difference between a country's exports (goods and services sold to other countries) and its imports (goods and services

bought from other countries). A positive net export balance means a country is exporting more than it is importing, while a negative balance means it is importing more than it is exporting.

GDP at factor cost by economic activity is adjusted by adding indirect taxes net of subsidies to arrive at estimated GDP at market prices so that it equals the expenditure on gross domestic product. The Gross National Product (GNP) at factor cost is arrived at by adding foreign income such as repatriated income, profits and royalties from abroad to the GDP at factor cost.

The Net National Product (NNP) is the net production of goods and services in a country during the year. It is simply the GNP adjusted for depreciation charges. The NNP gives a picture of the net increase in the total production of the country. It assists in the analysis of the long-run complexity of maintaining and enhancing the supply of physical capital in the country.

3.5 FLOW OF FUNDS ACCOUNTS

The Flow of Funds Accounts (FFA) is a macroeconomic tool which tracks the flow of funds within an economy. The FFA provides a widespread overview of the generation of funds and their distribution and utilization in an economy.

The FFA is concentrated by the Federal Reserve System of the United States, and it comprises statistics on certain sectors of the economy, such as households, businesses, government, banking and financial institutions and the rest of the world. There is confined tracking regarding the movement of funds between these sectors through varied financial instruments, like stocks, bonds, and loans.

The FFA is bifurcated into two main sections: the Asset and Liability section and the Transactions section. The Asset and Liability section gives detailed information on the financial assets and liabilities of each sector, while the Transactions section offers information on the financial transactions between the sectors.

Some of the instances of information offered by FFA are as follows:

- 1) Variation in the net worth of each sector.
- 2) The amount of credit outstanding in each sector.
- 3) The amount of equity outstanding in each sector.
- 4) The flow of funds between sectors, such as household savings being invested in business or government securities.
- 5) The composition of the financial assets and liabilities of each sector, such as the type of financial instruments held.

The FFA is an important macroeconomic tool for policymakers, researchers, analysts, and investors in recognizing the value of the financial structure of an economy and keeping track of the flow of funds between sectors. It can be used to explore potential areas of risk, monitor trends in investment patterns, and make policy judgements and decisions related to monetary and fiscal policy.

Apart from sector-wise information, the flow of funds is also organized along the financial instruments dimension. The macroeconomic analysis instrument-wise shows the aggregate preference of various sectors for different financial instruments. Financial claims are classified into two categories:

- Primary Issues or Direct Securities which incorporates all kinds of debt, marketable or issued by borrowers.
- Secondary Issues or Indirect Securities which incorporates debt and claims issued by financial intermediaries to acquire primary securities.

The total flow of finance is the sum of funds lifted by primary and secondary issues.

Flow of Funds-based Indicators of Financial Development

Goldsmith (1969) designed some ratios which contributed to understanding the role of financial development in the country. These ratios are termed financial deepening ratios.

Finance Ratio: This ratio indicates the financial structure of economic growth. This is defined as the total financial claims i.e., both primary issues and secondary issues divided by the national income.

$$FR = \frac{\text{Total Financial Claims}}{\text{National Income}}$$

Financial Inter-Relation Ratio (FIR): This ratio is the indicator of financial issues regarding capital formation of the economy. This demonstrates the relationship between financial structure and the real asset structure of an economy.

$$FIR = \frac{\text{Financial Assets (Total Issues)}}{\text{Physical Assets (Net Domestic Capital Formation)}}$$

New Issue Ratio (NIR): It is the proportion of primary issues to net domestic capital formation. This implies how primary or direct issues to the savers have financed the investment by the investing sectors.

$$NIR = \frac{\text{Primary Issues}}{\text{Net Physical Investments (Net Domestic Capital Formation)}}$$

Intermediation Ratio (IR): It is the indicator of secondary issues to primary issues and provides the significance of financial intermediaries in channelling

financial resources. It is the representation of financial claims issued by financial institutions and the financial instruments issued by non-financial institutions.

$$IR = \frac{\text{Total Secondary Issues}}{\text{Total Primary Issues}}$$

3.6 THE RELATIONSHIP BETWEEN STOCKS AND FLOWS

In accounting, stock implies the quantity or value of something that is measured at a specific point in time. Example: bank balance, inventory balance and debt owed by a company. On the other hand, flow implies the speed at which something is changing over some time. Example: the balance comes in and out of the bank, the rate at which the inventory is bought and sold or the revenue generation by the company.

Both are vital concepts from the accounting point of view. A company can gain insight regarding its financial position by keeping an eye on the stock of certain assets and liabilities. This helps the business firm or company to interpret how their financial position is wavering over time and can gain knowledge regarding the effectiveness of operations.

Stocks and flows play an important role in the Indian financial system. In financial terms, stocks refer to the current value of assets, while flows refer to the movement of funds or money in the economy. There are certain instances which show how financial system is impacted by stocks and flows:

- 1) *Investment*: Investment is highly impacted in the Indian financial system. Stock is known as the value of shares held by investors in a company. The high stock price of a company reflects that the company's prospects are good, and investors may show their readiness to invest in it. This leads to a rise in the flow of funds into the company, which can then be utilized for further investment.
- 2) *Capital markets*: The Indian capital markets are heavily impacted by stocks and flows. The stock market is the place where there exists trading of securities, and the flow of funds into the market ascertains the value of stocks. The heavy inflow of funds into the market enhances the demand for stocks thereby leading to an increase in stock prices. This, in turn, boosts the market value.
- 3) *Banking sector*: Stocks and flows also influenced the banking sector. A high inflow of funds into the economy leads the banks to lend more funds, ultimately increasing the supply of credit. This, in turn, leads to an increase in investment and economic growth.
- 4) *Foreign investment*: Stocks and flows positively attract foreign investment to India. When a high inflow of funds is generated into the economy, it signifies a robust economy thereby attracting foreign

investors. This would result in an increase in the value of stocks and hence, an increase in investment in the country.

Given the above, stocks and flows are essential elements of the Indian financial system. A proper balance of stocks and flows is necessary for sustained economic growth and development of the country.

Check Your Progress 2

- 1) Describe the macroeconomic tool which keeps track of the funds of an economy. Elaborate on its indicators of financial development.

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- 2) Illustrate the relationship between stock and flows. Explain the instances of how the financial system of an economy is impacted by the stock and flows.

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3.7 EXCHANGE RATES

Exchange rates are an important part of international financial markets. They express the value of one currency to another. Exchange rates have a huge impact on the economy, including gross domestic product (GDP) and government debt. In this section, we will examine the history of exchange rates, their role in international financial markets, and the various factors that affect exchange rates. The history of exchange rates goes back to the early days of commerce when people began to exchange goods and services.

However, it wasn't until the 19th century that exchange rates began to work. The gold standard was introduced in 1875 and fixed its value in gold. The system existed until the First World War when the country began to abandon it due to financial constraints.

After the Second World War, the Bretton Woods system was established to fix the value of US gold and other currencies to the US dollar. However, in the early 1970s, the system collapsed due to the economic crisis and the US's failure to maintain the gold standard.

Today, most countries have a floating exchange rate, meaning the value of their currencies is determined by the market forces of supply and demand. The system is more flexible and less susceptible to financial losses.

Exchange rates are related to international financial markets in many ways. For example, the exchange rate affects the value of imports and exports, which affects a country's balance of payments. Exchange rates also affect the value of a country's assets, such as stocks and bonds. Changes in exchange rates can cause significant changes in financial markets and investors should take this into account in their investment decisions.

One of the most important effects of the exchange rate is on GDP. When a country's currency strengthens, its exports become more expensive, resulting in fewer exports and lower GDP. Conversely, when a country's currency is weak, its exports become cheaper, leading to higher exports and higher GDP.

For example, let's take the exchange rate between the US dollar and the Euro. The exchange rate of 1 USD = 0.85 EUR. If the dollar strengthens against the euro, the exchange rate will change to \$1 = €0.80. This will make U.S. exports more expensive to European countries, which could lead to lower U.S. exports and lower GDP.

Government borrowing is also affected by exchange rates. When the country's currency weakens, it becomes more expensive for the government to borrow from foreign investors. This can lead to higher borrowing costs and lower government spending. Conversely, when a country's currency is strong, it becomes cheaper for the government to borrow money, leading to lower borrowing costs and more government spending.

For example, consider the exchange rate between the Japanese Yen and the US Dollar.

Suppose the exchange rate is 110 Yen = \$1. If the yen weakens against the dollar, the exchange rate will change to 120 yen = \$1. This would make it more expensive for the Japanese government to borrow money.

In summary, exchange rates play an important role in international financial markets. Understanding the history and factors that affect exchange rates is important for investors and policymakers. The exchange rate affects GDP, government borrowing and the financial market, and changes in the exchange rate have a significant impact on the economy. By monitoring exchange rates and taking steps to contain their effects, countries can help ensure stability and growth.

3.8 RATE OF INTEREST

Interest rates are an important part of today's economy. It refers to the cost of borrowing used by individuals, businesses and governments to finance investment and business. In this section, we will examine the history of interest, its post-exchange rate policy and its impact on the economy.

Trading by barter before interest is charged. Under this system, goods and services can be exchanged for other goods and services without spending money.

However, the barter system has its limitations. For example, it is difficult to assess the value of goods and services, and it is not easy to find work to exchange goods or services with anyone.

With the emergence of money, the concept of interest emerged. The interest rate reflects the cost of the loan and is usually expressed as a percentage of income. Interest rates allow individuals, businesses and governments to borrow money to invest and operate and provide a way for borrowers to get a return on their investment.

An example of the use of interest is in the real estate sector. When a person wants to buy a house but does not have the cash to pay, they can apply for a loan. A mortgage is a cost-of-purchase loan and is usually repaid with interest for 15 to 30 years. The interest rate on your mortgage can have a big impact on the cost of the loan and the affordability of your home.

Governments can borrow money mainly by issuing promissory notes that promise to repay the money and interest at a certain rate. Interest rates on government bonds are determined by economic forces that reflect the government's borrowing risk. If investors perceive that the government is a risky borrower, they will charge higher interest rates to cover the risk.

Interest rates also affect the economy as a whole. When interest rates are high, it discourages lending and investment, which slows the economy. Conversely, when interest rates are low, it encourages lending and investment, which stimulates economic growth.

As was witnessed after the 2008 financial crisis, the central banks around the world lowered interest rates to encourage lending and investment. For example, the Federal Reserve lowered the cost of borrowing for businesses and individuals by lowering the federal funds rate to zero. This contributes to the growth of the economy and prevents further declines.

Interest rates were introduced after the barter system as a way to measure the value of the loan. Interest rate is used by individuals, businesses and governments to finance investment and business and provides a way for borrowers to get a return on their investment. Interest rates can affect the entire economy, and their use is monitored by policymakers and investors.

3.9 GOVERNMENT BORROWINGS

Government borrowing is the government's process of raising money by issuing bonds or other securities to investors. These funds are used to finance government expenditures such as infrastructure, health services and defence. In this section, we will examine the government borrowing process, how it is used to illuminate the impact of economic policy, and the role of the family in the economy.

Investors such as banks, pension funds, and individual investors purchase government securities and lend money to the government. The government then uses the proceeds of these securities to fund its expenditures.

One way to use government debt to illustrate the impact of monetary policy on the economy is the concept of government deficit. The government's budget deficit is the difference between government expenditures and revenues in a year. If the government spends more than necessary, it runs a budget deficit that must be financed by borrowing.

The size of the government's budget deficit can have a significant impact on the economy. When the government borrows to finance its spending, they compete with private lenders for money. This could lead to higher interest rates affecting the private sector and slowing the economy. On the other hand, if the government borrows less and spends more, it can help lower interest rates and stimulate private investment, which stimulates economic growth.

Another way government loans affect the economy is through the role of the households. When the government borrows to finance its spending, it pulls more people out of the private sector, which means households have less money to invest. This affects the wealth of households and their consumption patterns, which in turn affects the economy.

For example, if households have less money to invest, it will slow economic growth by reducing their consumption of goods and services. Or, if households have less money to invest, they may choose to invest in risky assets to make more money. This can lead to asset bubbles and financial instability.

3.10 NOMINAL AND REAL INTEREST RATES

The nominal interest rate refers to the accounting rate of interest stated on a particular loan or investment. It does not take into account the impact of inflation on the purchasing power of money. For example, if a bank offers a 10% interest rate on a savings account, the account holder will earn 10% interest on their deposits, regardless of the inflation rate.

Nominal interest rates can be influenced by various factors, including the demand and supply of money, the actions of the central government, the monetary policy of the central bank, and many others. Central banks implement short-term nominal interest rates as a monetary policy. The nominal rate is usually decreased to stimulate economic activity during recessions. However, the nominal rate increases during the inflationary phase.

On the other hand, the real interest rate is adjusted for inflation and represents the true return on investment after accounting for inflation. The real interest rate equals inflation less the nominal interest rate. For example, if the nominal interest rate is 10% and the inflation rate is 4%, the real interest rate would be 6% (10% - 4%).

Real interest rates are particularly significant in financial markets as they provide a more accurate measure of return on investment. A high nominal interest rate may appear attractive, but if the inflation rate is high, it can erode the purchasing power of the investment. Therefore, investors often look at real interest rates when evaluating investment opportunities.

In addition, real interest rates can also influence credit supply and demand in financial markets. For instance, when real interest rates are low, borrowers may be more likely to take out loans to finance their investments. This may lead to an increase in credit demand. Conversely, when real interest rates are high, borrowers may be less inclined to borrow, which can lead to a decrease in credit demand.

Overall, both nominal and real interest rates are critical concepts in finance. Understanding their differences and their impact on financial markets is crucial for investors, borrowers, and lenders alike.

Inflation Adjustment of the Nominal Interest Rate:

The Fisher Effect explains the relationship between inflation and nominal or real interest rates through the equation below:

$$(1 + i) = (1 + R) (1 + h)$$

Where:

i – Nominal interest rate

R – Real interest rate

h – Expected inflation rate

In a stable economy that grows at a moderate pace, the inflation rate is usually low. With a low inflation rate, a simplified Fisher equation can be implemented. According to this formula, the nominal interest rate is equal to the real interest rate plus inflation ($i = R + h$).

A bond investor, for example, expects a real interest rate of 10%, when the market expects inflation of 4%. For this reason, investors should search for bonds with stated (nominal) interest rates of 14% ($10\% + 4\%$).

Using the formula below, you can calculate the nominal interest rate.

$$i = (1 + R) (1 + h) - 1 = (1 + 10\%) (1 + 4\%) - 1 = 14.4\%$$

According to the Fisher effect, if the nominal interest rate remains constant and the inflation rate rises, the real interest rate falls. The lower the purchase price, the better the lender. If both the nominal interest rate and the expected money supply increase at the same rate, that is, if the nominal interest rate increases, the real interest rate does not change.

Check Your Progress 3

- 1) Discuss the importance of exchange rates in the international financial markets.

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- 2) What are government borrowings? How government borrowings are financed through banks?

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- 3) What is meant by nominal and real interest rates? Explain the relationship between inflation and nominal or real interest rates.

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3.11 LET US SUM UP

In this unit, basic accounting concepts are explained in the context of the financial system. There has been an emphasis on financial statements such as stocks and flows. Stock is known by the balance sheet and flow statements are classified as income statements and cash flow statements. Real assets and financial assets are also being discussed. Both have their features and the relationship between them is complicated as well as dynamic. Their disclosure is also different in the balance sheet. Further, the focus is on the national income accounts which helps in measuring the economic performance of the country. Thereafter, the flow of flows being the macro-economic tool is elaborated. This tracks the flow of funds in an economy. There are several flows of funds-based indicators of financial development. Certain instances have also been explained demonstrating the impact of stock and flows on financial development. Exchange rates also play an important role in the international financial markets. The history of interest rates, its

policy related to exchange rates and its impact have also been focussed upon. There is further the role of government borrowings being explained and how it is influencing the economic policy. Lastly, the nominal and real interest rates have been explained which are important.

3.12 KEY WORDS

- Barter System** : any exchange of goods and services for other goods and services with no exchange of money.
- GAAP** : Collection of commonly followed accounting rules and standards for financial reporting, known as generally accepted accounting principles.
- Intermediaries** : Individuals or companies who work as middlemen or act as a link between parties for investment deals, business deals, negotiations, insurance, etc.,
- Monetary Policy** : It is a set of tools used by the central bank to control the money supply and promote economic growth in the economy.

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3.14 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) Read Section 3.2
- 2) Read Section 3.3

Check Your Progress 2

- 1) Read Section 3.5
- 2) Read Section 3.6

Check Your Progress 3

- 1) Read Section 3.7
- 2) Read Sections 3.8 and 3.9
- 3) Read Section 3.10



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